Borrowing History Data Analysis

August 2024 – August 2026

LIBRARY DATA TALK

Xiamen University Malaysia

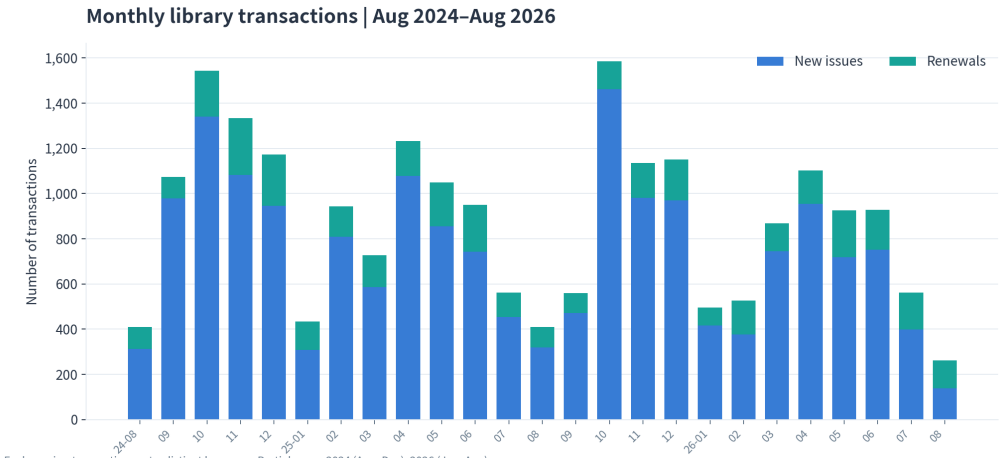
9 October 2026

Dataset at a glance

Transactions	New issues	Renewals
21,943	18,196	3,747

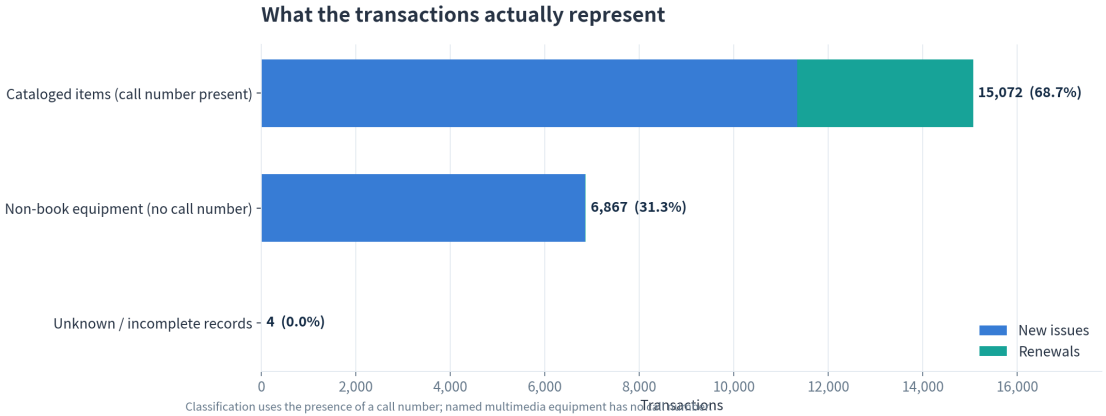
- **Period:** 1 Aug 2024 – 30 Aug 2026; 25 consecutive months.
- **Unit of observation:** a recorded issue or renewal, not a unique borrower.
- **Seven fields:** title, author, barcode, call number, edition, date/time and transaction type.
- **Scope:** the export includes classroom multimedia equipment as well as cataloged materials.

Trend 1 | Monthly circulation



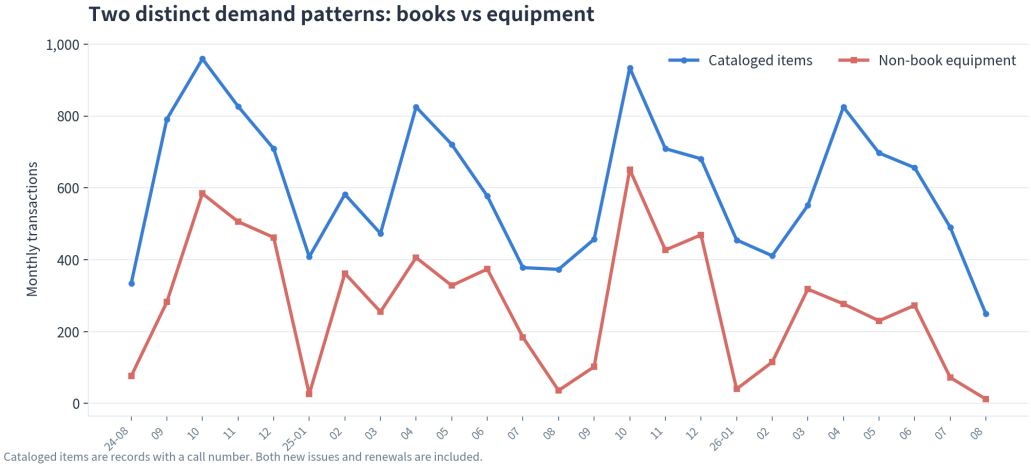
October has substantial activity in both 2024 and 2025; partial-year totals must not be treated as full-year growth.

Finding 1 | Materials are not all books



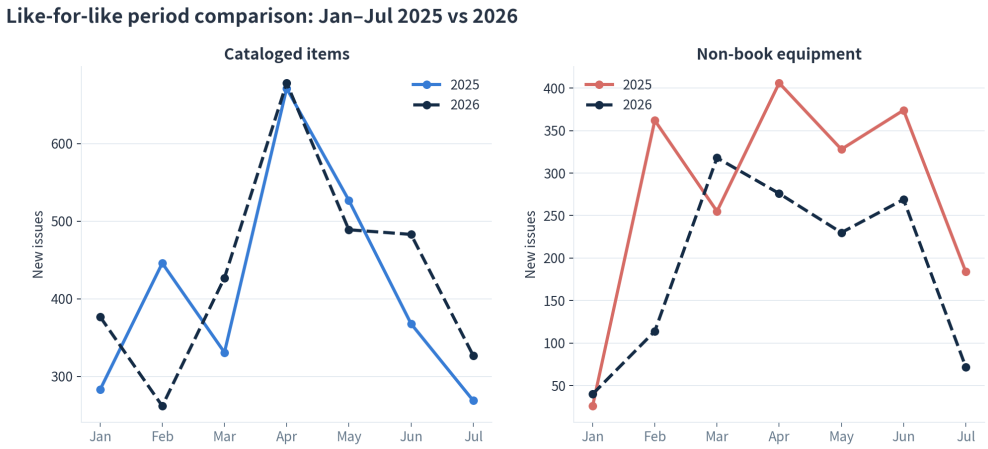
Access cards, microphones, pointers and other equipment account for a large share of all transactions.

Finding 2 | Two demand patterns



Cataloged borrowing and classroom equipment usage should be evaluated separately.

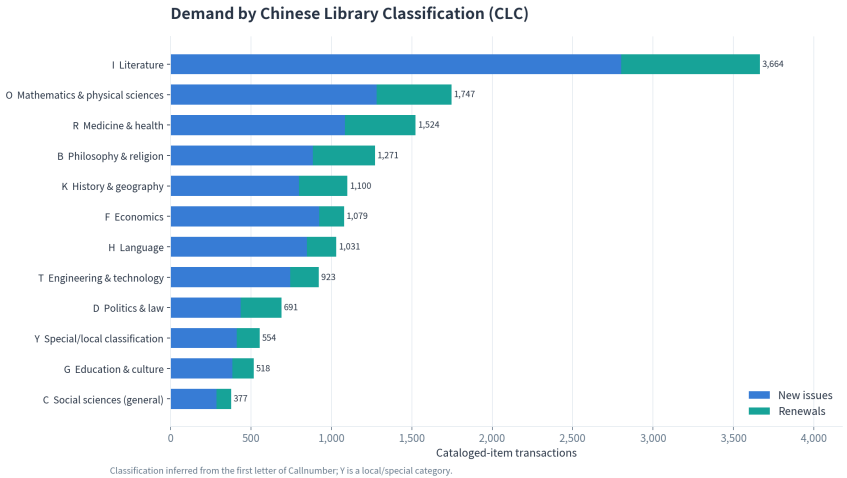
Like-for-like comparison | Jan–Jul



Monthly NEW ISSUES only, Jan–Jul (same months in both years). Descriptive differences are not causal growth estimates.

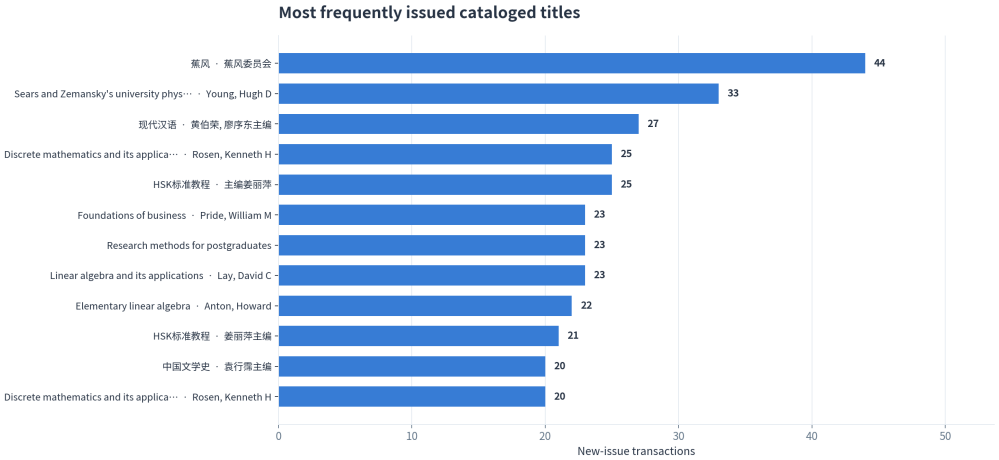
Only new issues, matched month-for-month in 2025 and 2026; descriptive comparison rather than a causal claim.

Which subjects receive the most demand?



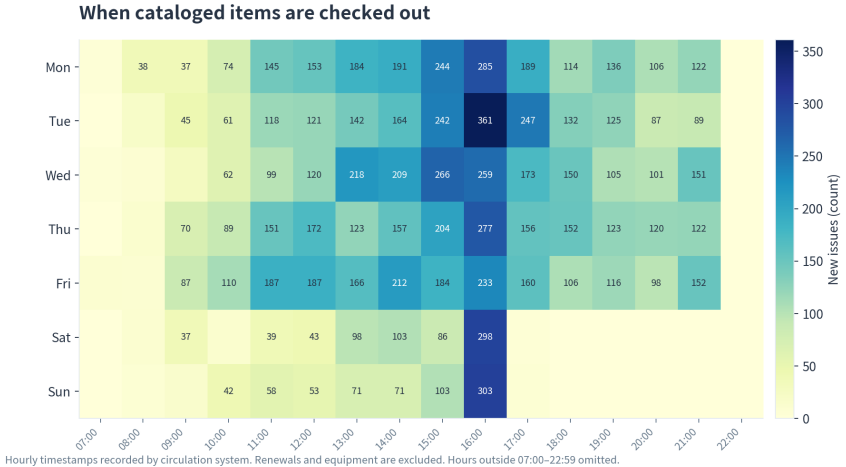
Subject groups derive from the first letter of the library call number (CLC); counts include issues and renewals.

Frequently issued cataloged titles



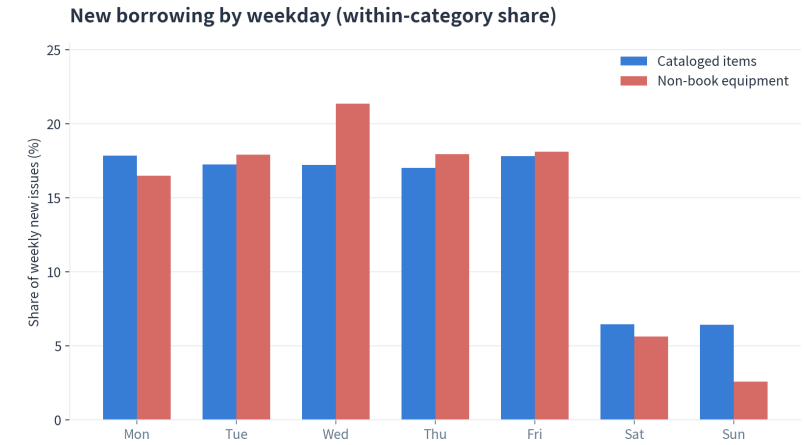
Title + Author grouping prevents some collisions from generic titles such as “Calculus”; magazines can still appear.

Borrowing rhythm | Hour and weekday



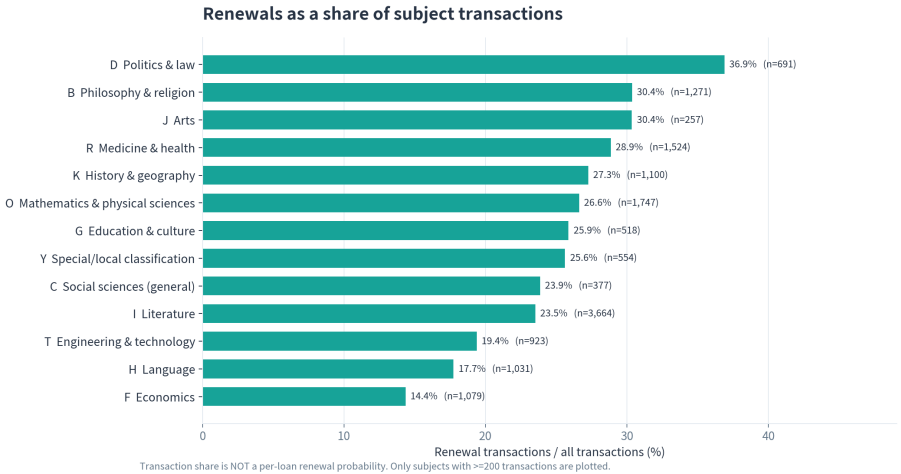
New issues of cataloged items only; timestamp does not reveal when a reader actually used the item.

Borrowing rhythm | Compare resource types



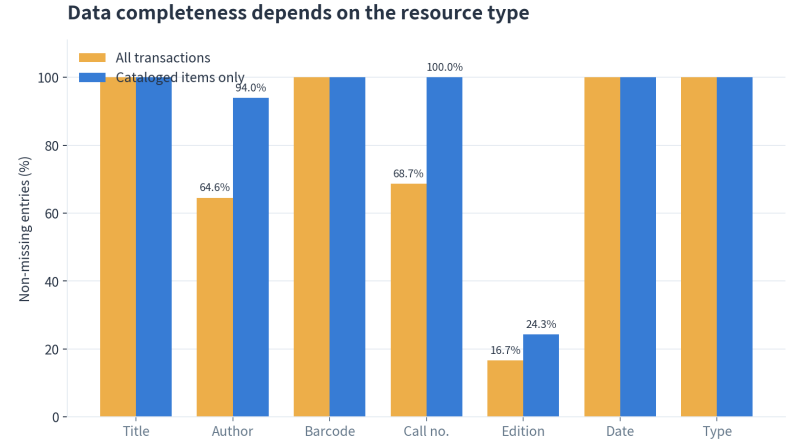
Normalized weekly shares highlight how classroom equipment and cataloged materials follow different schedules.

Renewal patterns by subject



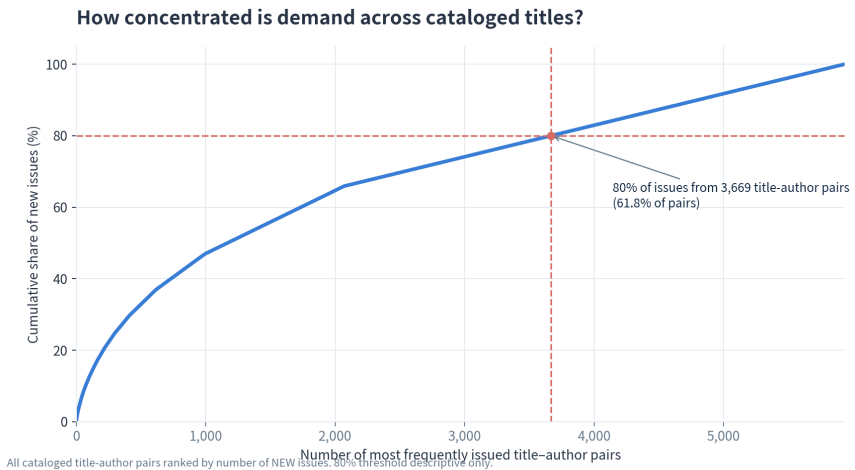
Renewal-to-transaction ratio is an activity proxy, not the probability of a loan being renewed.

Data quality | Completeness by field



Equipment lacks call numbers by design; edition information is often not populated even for cataloged items.

Collection usage | Long tail



Borrowing is distributed across many titles; popularity should be considered together with the number of available copies.

Recommendations and cautions

Actions suggested by the data

- Separate **cataloged materials** and **multimedia equipment** in future dashboards.
- Track **subject-level circulation** to identify areas for collection review.
- Use **weekday and hour patterns** for operational staffing discussions.
- Compare months like-for-like instead of comparing incomplete years.

What this dataset cannot establish:

- Number of unique borrowers, unmet demand, return delays, or loan duration.
- Whether a popular title is under-supplied without copy availability data.
- Why seasonal changes happened without a matching academic calendar.

Methodology | Reproducible definitions

1. Use the supplied XLSX transaction export; treat each row as an event.
2. Parse the circulation timestamp and retain `issue` / `renew` separately.
3. Define cataloged material as a row with a non-blank call number; classify the identifiable equipment records separately.
4. Derive a subject code from the initial character of each call number.
5. Show transaction counts, never reinterpret renewals as distinct loans or people.
6. Retain exact repeated records pending source-system validation.

Limitations: five identical rows; four records without title/barcode/call number; no patron IDs, loan returns or due dates.
Data span: 2024-08-01 to 2026-08-30. Original ODS also supplied; analysis reads the XLSX export.

Thank you

Questions & discussion